RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Thursday, 27 August 2026, 11:35 AM
State	Finished
Completed on	Sunday, 30 August 2026, 8:48 PM
Time taken	3 days 9 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```
 2
 3  void swap(int *a, int *b) {
 4      int temp = *a;
 5      *a = *b;
 6      *b = temp;
 7  }
 8
 9  int partition(int arr[], int low, int high) {
10      int pivot = arr[high];
11      int i = low - 1;
12
13      for (int j = low; j < high; j++) {
14          if (arr[j] <= pivot) {
15              i++;
16              swap(&arr[i], &arr[j]);
17          }
18      }
19
20      swap(&arr[i + 1], &arr[high]);
21      return i + 1;
22  }
23
24  void quickSort(int arr[], int low, int high) {
25      if (low < high) {
26          int pi = partition(arr, low, high);
27
28          quickSort(arr, low, pi - 1);
29          quickSort(arr, pi + 1, high);
30      }
31  }
32
33  int main() {
34      int n;
35      scanf("%d", &n);
36
37      int arr[n];
38
39      for (int i = 0; i < n; i++) {
40          scanf("%d", &arr[i]);
41      }
42
43      quickSort(arr, 0, n - 1);
44
45      for (int i = 0; i < n; i++) {
46          if (i > 0)
47              printf(" ");
```

```
48          printf("%d", arr[i]);
49      }
50
51      return 0;
52  }
53
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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